

IIIT-PGEE



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 01



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Topics to be Covered



Topic-1

Calendar ✓

Topic-2

Clocks ✓



CALENDAR



✓ The cycle of leap years

✓ Theory of odd days

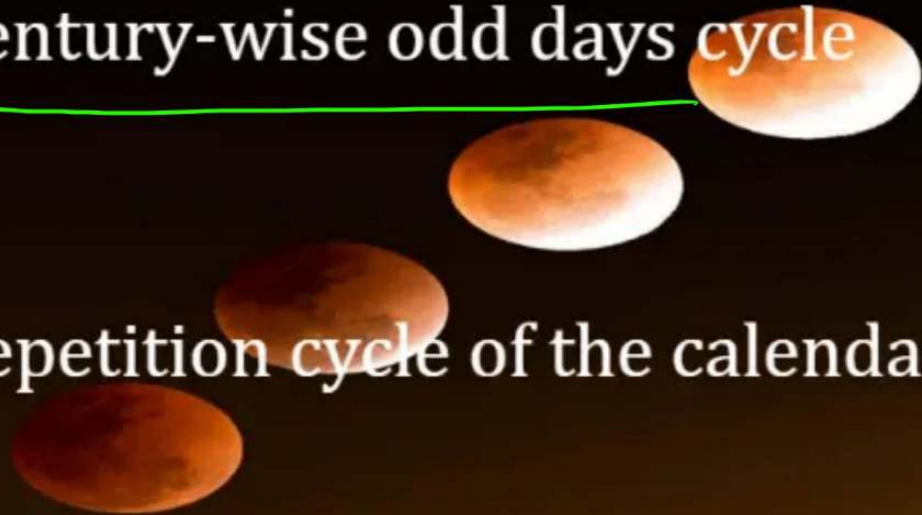
✓ Century-wise odd days cycle

✓ Repetition cycle of the calendar

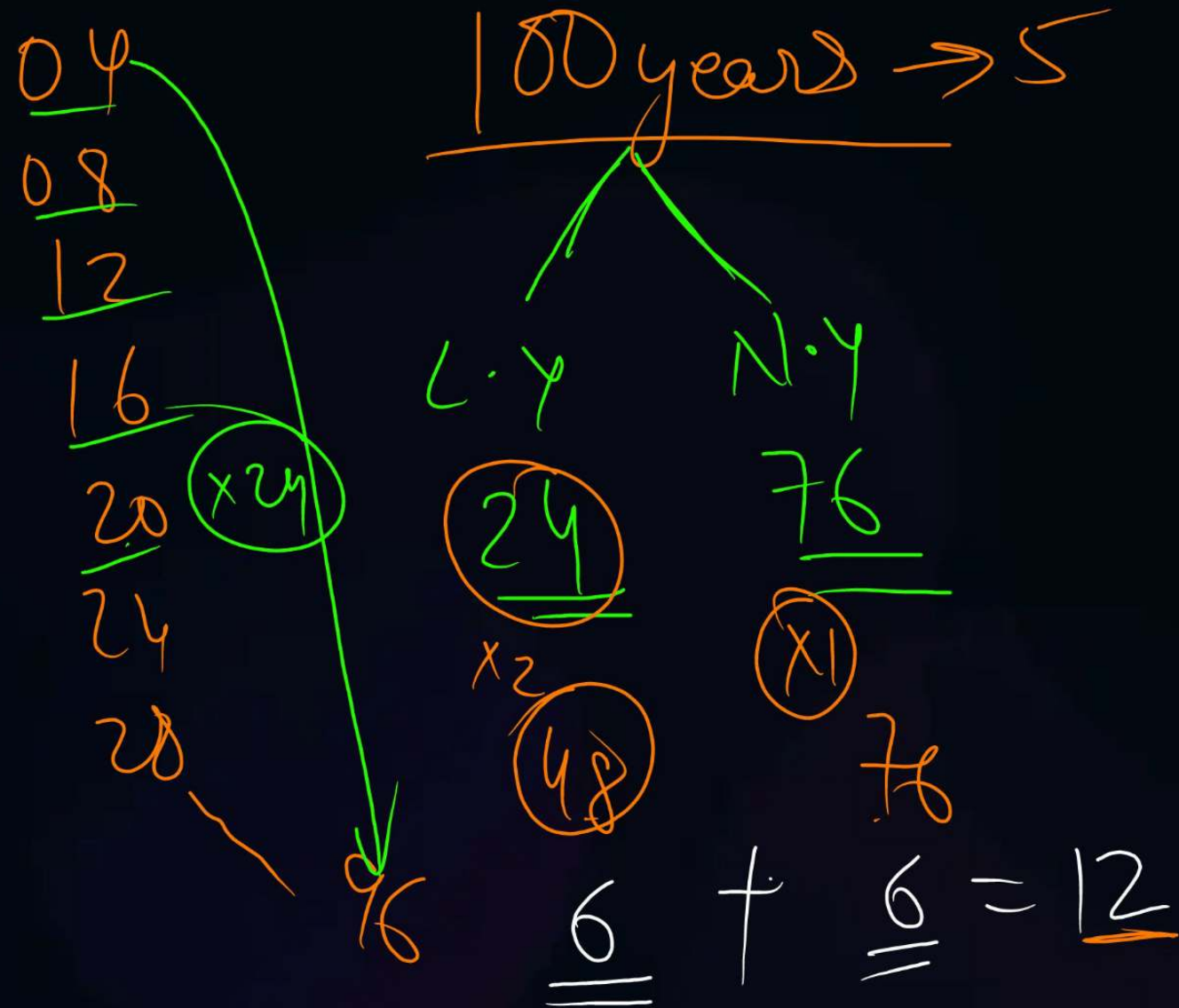
⑦) $\frac{69}{63}$ (9 weeks)
odd days
69 days
Century
1652
400
L.Y



1692
1696
4 yrs
8 yrs
? 1700 x 1704



ODD DAYS



N.Y. $\rightarrow$ 365 days $\rightarrow$ 1 odd day

L.Y. $\rightarrow$ 366 days $\rightarrow$ 2 odd days

52 weeks

500 100 $\rightarrow 5$

600 200 $\rightarrow 3$ (10)

700 300 $\rightarrow 1$ (15)

800 400 $\rightarrow 0$ (20+1)

Odd Days in Centuries



SUNDAY
 $2 + 5 = 7$

→ Odd day

12th October, 2003

2002

$$2000 + 2$$

$$0 + 2 = \underline{2}^{\text{odd}}$$

A - 2 J - 3 O - 5
 M - 3 + A - 3 O - 5
 J - 2 S - 2
5

1300	900	500
1400	1000	600
1500	1100	700
1600	1200	800

100 - 5
200 - 3
300 - 1
<u>400 - 0</u>

- 0 - Sun
- 1 - Mon
- 2 - Tue
- 3 - Wed
- 4 - Thu
- 5 - Fri
- 6 - Sat
- 7 - 3

Questionnaire:

$$\begin{array}{c}
 6 \\
 \swarrow \searrow \\
 L-4 \quad N-4 \\
 1 \quad 5 \\
 \textcircled{2} + \textcircled{5} = 7
 \end{array}$$

22nd Jan, 2007 ↗ odd day

$$\begin{array}{c}
 2000 + 6 \\
 \downarrow \quad \downarrow \\
 0 \quad 0 \\
 \text{---} \\
 0 \\
 \text{---}
 \end{array}
 \quad J \Rightarrow 1$$

- ☒ (A) Monday
- (B) Sunday
- (C) Tuesday
- (D) None

Questionnaire:

#Q. On which day of the week does 15th March, 2040 fall?

$$2039 \rightarrow 0 + 6 = \underline{\underline{6}}$$

Thursday $\rightarrow$ 4 odd days
 $6 + 5 = \underline{\underline{11}}$

04
 $\times 9$
36

A Sunday

B Tuesday

C Thursday

D Saturday

$$\begin{array}{r} 2000 \\ \hline \downarrow \\ 0 \end{array} + 39 \Rightarrow \underline{\underline{6}}$$

0

L.Y.

N.Y.

9

30

18

30

4

+ 2 = 6

J-3

F-1

M-1 (15)

5

Calendar Rep

	^{N.Y.} <u>2026</u> →	1	^{N.Y.} <u>2037</u> ✓
2032 →	2		
2033 →	1		
2034 →	1		
2035 →	1		
<u>2036 →</u>	<u>2</u>		
	2027 →	1	
	2028 →	2	
	2029 →	1	
	2030 →	1	
	<u>2031 →</u>	<u>1</u>	

Questionnaire:

#Q. 2001 calendar is same as which of the given years?

- A** 2009
- B** 2012
- C** 2011
- D** 2007

^{N.Y}
2001 → 1
 2002 → 1
 2003 → 1
 2004 → 2

^{N.Y}
2007

2005 → 1
 2006 → 1

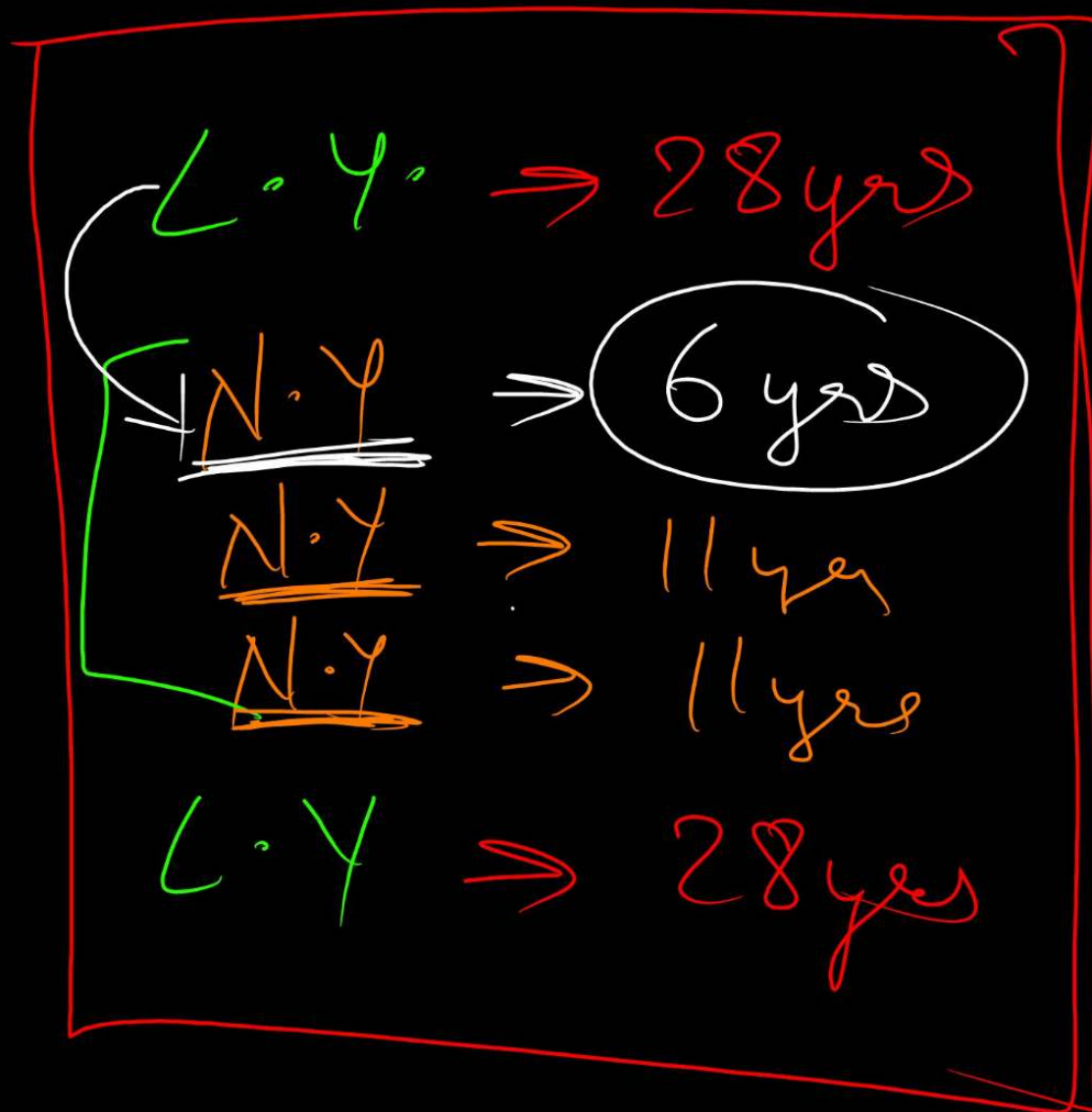
$$\begin{array}{r} 2013^u \\ + 6 \\ \hline 2019^u \\ 2032^u \end{array}$$

$$\begin{array}{r} 2032^u \\ + 28 \\ \hline 2060^u \end{array}$$

1899

1900 X

$$\begin{array}{r} 2004^u \\ + 28 \\ \hline 2032^u \end{array}$$



1997

2000

$$\begin{array}{r} 2009^u \\ + 6 \\ \hline 2015^u \end{array}$$

[MCQ]



#Q. 2022 calendar is same as which of the given years?

2024

+11

2033

A

2033

B

2047

C

2039

D

2035

[MCQ]



#Q. When does 2012 calendar repeats?

+28
2040

A 2048

B 2040

C 2020

D 2044

[MCQ]



#Q. Netaji Subhas Chandra Bose was born on 23rd January 1897. What was the week Day?

- A** Monday
- B** Thursday
- C** Tuesday
- D** Saturday

$$1896 \rightarrow 4$$

$$J \Rightarrow 2 (23)$$

$$6$$

$$1800 + 96 \Rightarrow 1$$

↓

3

$$\begin{array}{l} L: 4 \quad N: 4 \\ 24 \quad 72 \\ 48 \quad 72 \\ \hline 6 + 2 = 8 \end{array}$$

24
x 24
96

[MCQ]



#Q. If September 10th, 1982 was a Wednesday. What day of the week was October 2nd, 1984?

+4 odd days

10th Sep 1982 — 2nd Oct 1984

10th Sep 1982 — 10th Sep 1984

Sep $\rightarrow 6(20)$

Oct $\rightarrow 2$

8 $\rightarrow 1$

2 yr $\rightarrow 3$

N.Y. ① L.Y. ①

① + ② = 3

A

Sunday

B

Wednesday

C

Thursday

D

Friday

Questionnaire:



1st Feb → Monday
29th Feb → Monday
Feb → 29

#Q. ✓ A person was born on fifth Monday of February in a particular year.
Which one of the following statements is correct based on the above information?.

- ☒ **A** The 2nd February of that year is a Tuesday.
- ☐ **B** There will be five Sundays in the month of February in that year.
- ☐ **C** The 1st February of that year is a Sunday.
- ☐ **D** All Mondays of February in that year have even dates.

Questionnaire:

#Q. Which day of the week can the final day of a century fall on?

- A** Tuesday
- B** Wednesday ✓
- C** Thursday
- D** Saturday

100	→ 5	] <u>Friday</u>
200	→ 3	
300	→ 1	
400	→ 0	
		<u>Wednesday</u>
		<u>Monday</u>
		<u>Sunday</u>

CLOCK



Relative Speed:



5.5°/minute $\frac{\text{Time}}{\text{Angle}}$ $\rightarrow$ Angle? $\rightarrow$ Time?

opposite $\rightarrow$

Same $\rightarrow$ -

per minute

$$H.H \rightarrow \frac{360^\circ}{12 \times 60} = \frac{1^\circ}{2}$$

$$= \frac{1^\circ}{2} \text{ minute}$$

$$M.H = \frac{360^\circ}{60} \text{ per minute} = \underline{6^\circ/\text{minute}}$$

$$H.H \rightarrow 0.5^\circ/\text{min}$$

$$M.H \rightarrow \underline{6^\circ/\text{minute}}$$

To be Noted:

✓ Minute Hand Covers 30° in 5 minutes $30^\circ/5 =$

$6^\circ/\text{min.}$ ✓

✓ Hour Hand Covers 30° in 60 minutes $30^\circ/60$

$= \frac{1}{2}^\circ / \text{min}$
or

$0.5^\circ/\text{min}$ ✓

5.50/minute

Different Patterns of Questions:

Time $\rightarrow$ Angle?

Angle $\rightarrow$ Time?

Gain OR loss

First Pattern:

$$\frac{360^\circ}{12} = 30^\circ$$

5:00 $\rightarrow 150^\circ$

8 $\rightarrow 240^\circ$

2:00 $\rightarrow 60^\circ$

10 $\rightarrow 300^\circ$

7:00 $\rightarrow 210^\circ$



Random time given:

11:23

$$11 \rightarrow 330^\circ$$

$$23 \times 5.5 \rightarrow 126.5^\circ$$

$$\underline{\underline{203.5^\circ}}$$

7:45

$$\rightarrow 37.5^\circ \text{w}$$

$$7 \rightarrow 210^\circ$$

$$45 \times 5.5 \rightarrow 247.5^\circ$$

$$\underline{\underline{37.5^\circ}}$$

Questionnaire:

- (A) 67.5°
- (B) 330°
- (C) 22.5°
- (D) 180°

4:40

$$4 \Rightarrow 120^\circ$$

$$40 \times 5.5 \Rightarrow 220^\circ$$

$$\underline{\underline{180^\circ}}$$



[MCQ]



#Q. At 7:45, the two hands of a clock make an angle of?

A

60°

B

45°

C

$33 \frac{1}{3}^\circ$

D

$37 \frac{1}{2}^\circ$

$$7 \Rightarrow 210^\circ$$

$$45 \times 5.5 \Rightarrow 247.5^\circ$$

$$\underline{37.5^\circ}$$

[MCQ]

#Q. At 2:15, the two hands of a clock make an angle of?

Assignment



- A** 22.5°
- B** 15°
- C** $16 \frac{1}{3}^\circ$
- D** $7 \frac{1}{2}^\circ$

Second Pattern:

Angle [✓] $\rightarrow$ Time?





2 mins Summary



Topics

Calendar & Clocks





THANK - YOU

